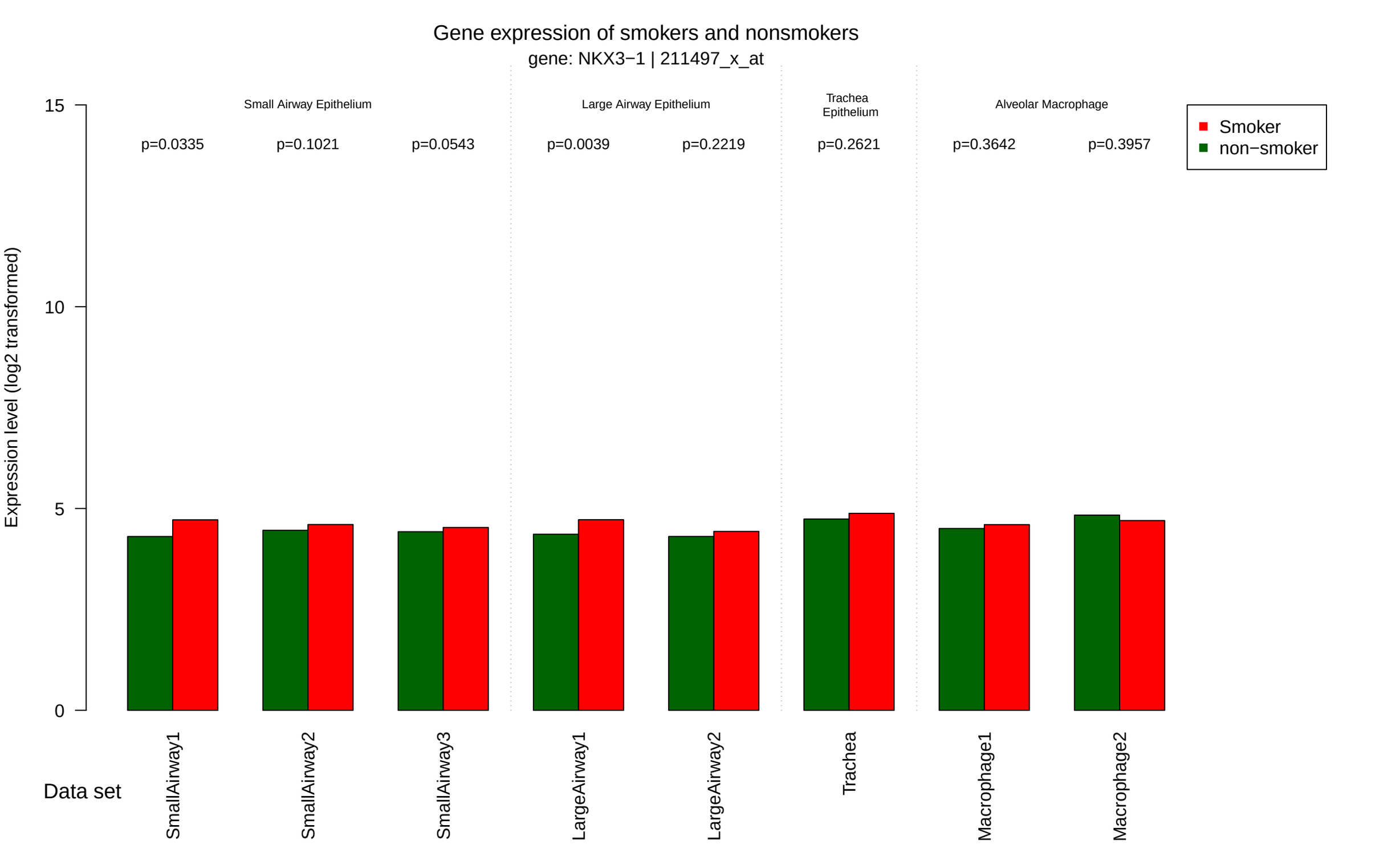
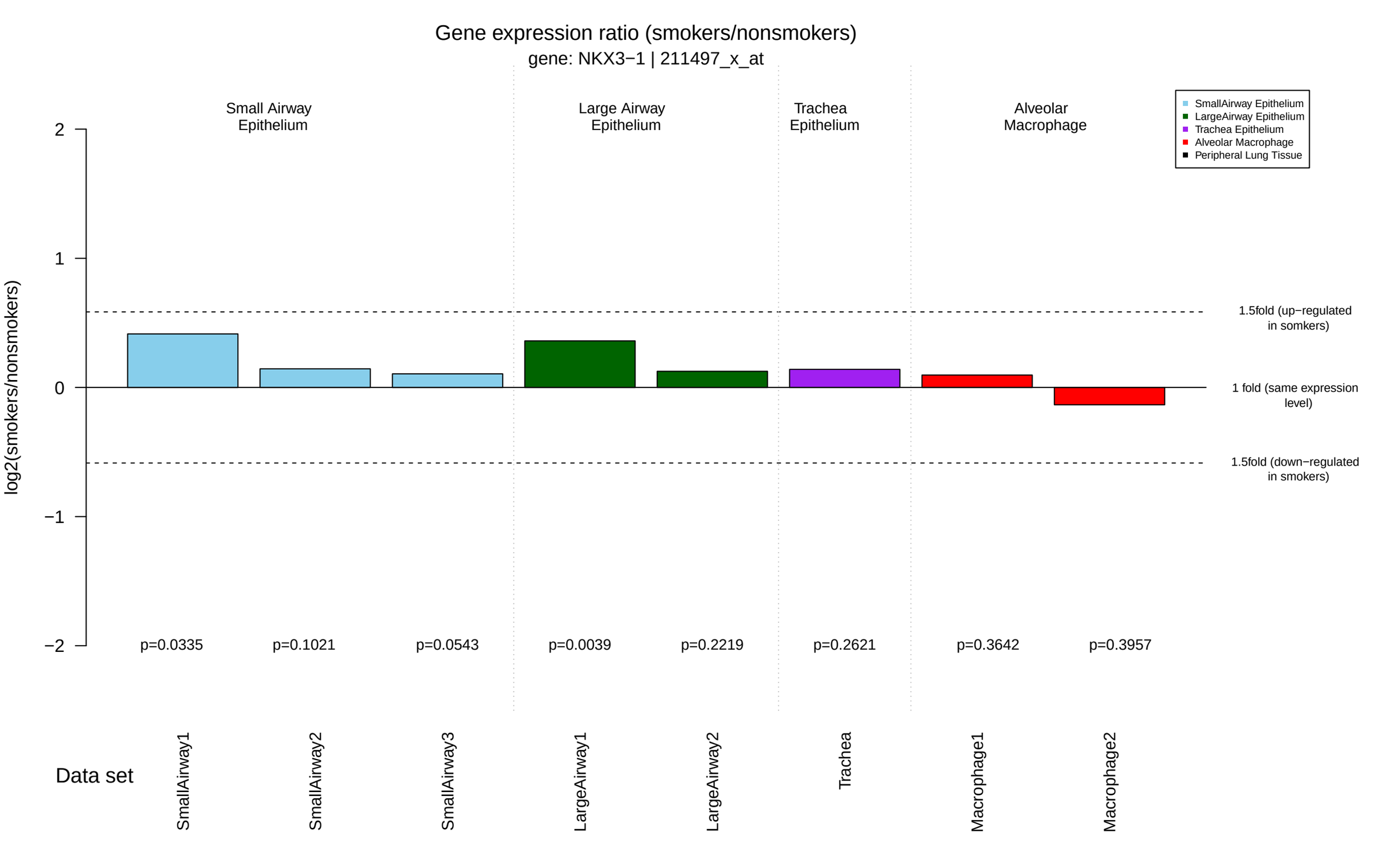
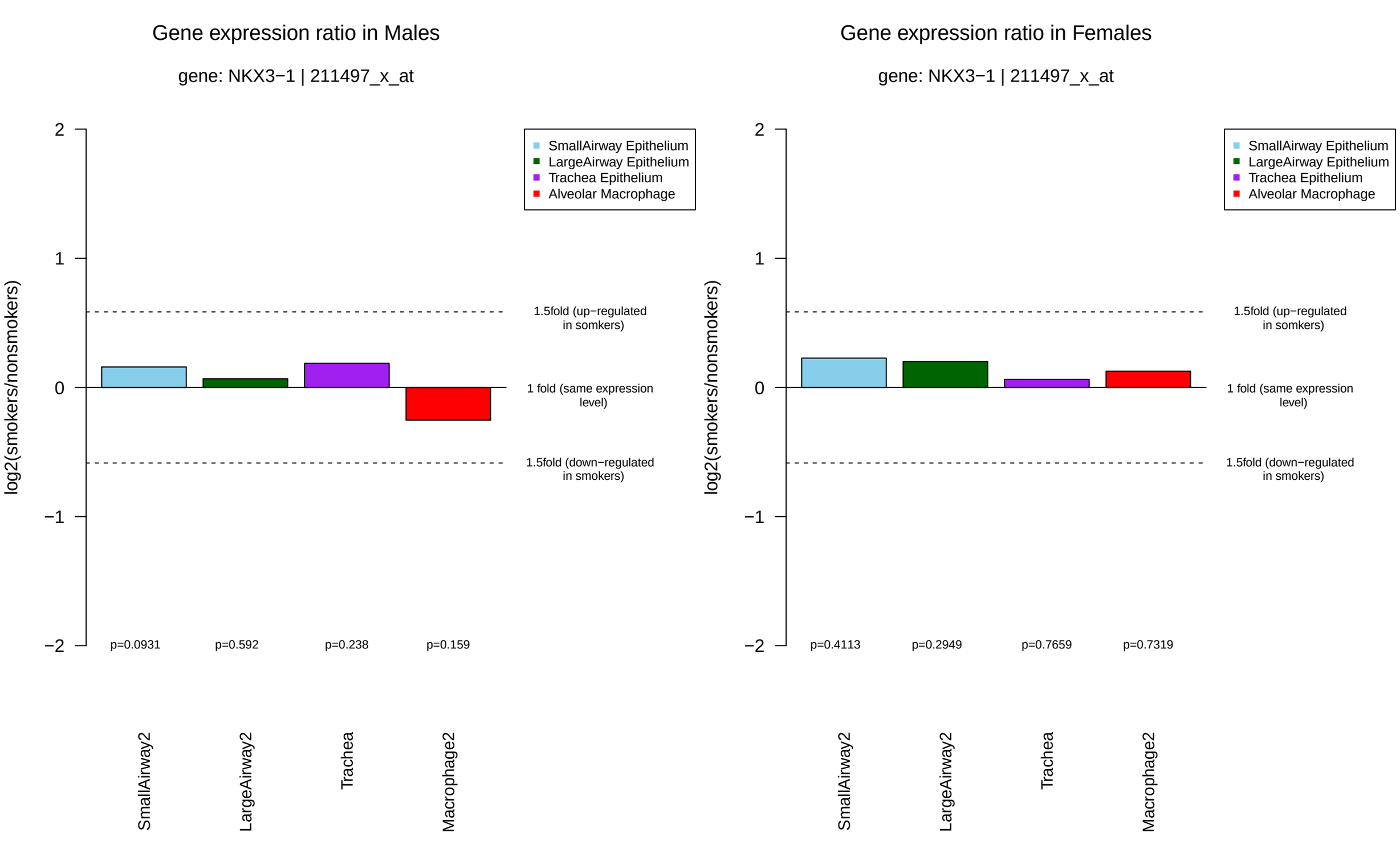
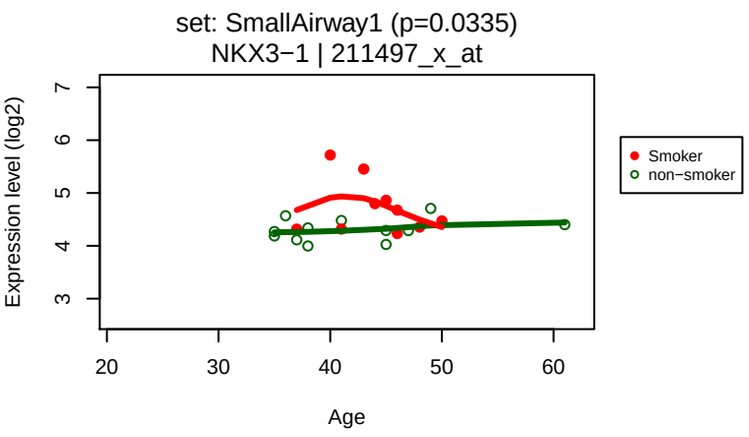
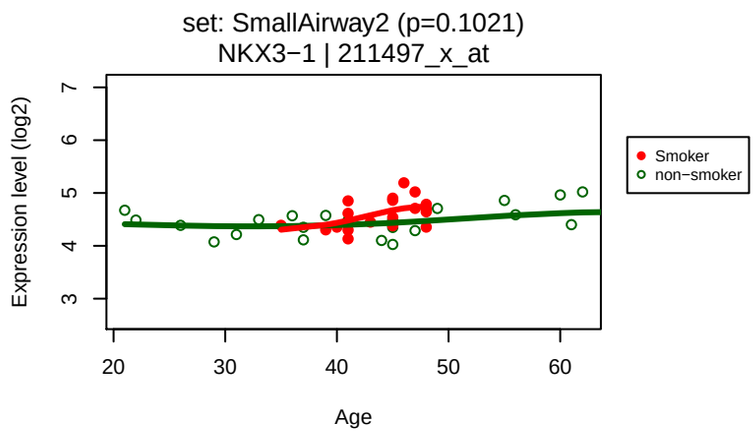
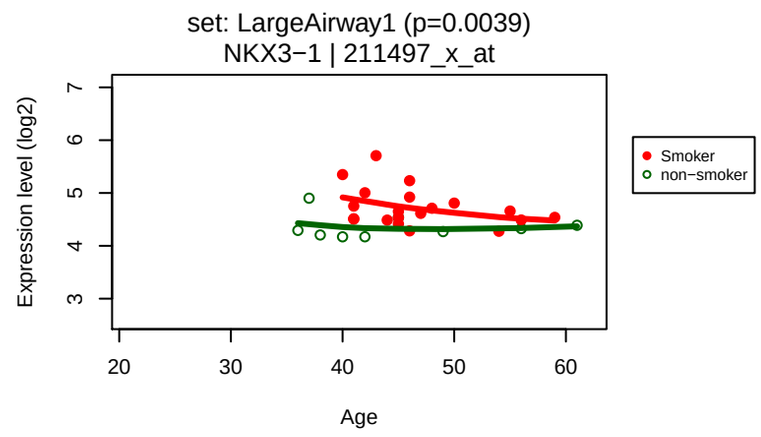

Gene expression of smokers and nonsmokers
gene: NKX3−1 | 211497_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0335
p=0.1021
p=0.0543
p=0.0039
p=0.2219
p=0.2621
p=0.3642
p=0.3957
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

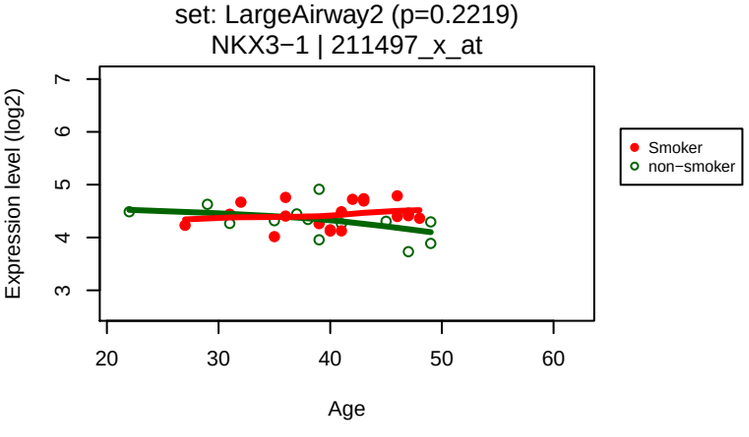

Gene expression ratio (smokers/nonsmokers)
gene: NKX3−1 | 211497_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0335
p=0.1021
p=0.0543
p=0.0039
p=0.2219
p=0.2621
p=0.3642
p=0.3957
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

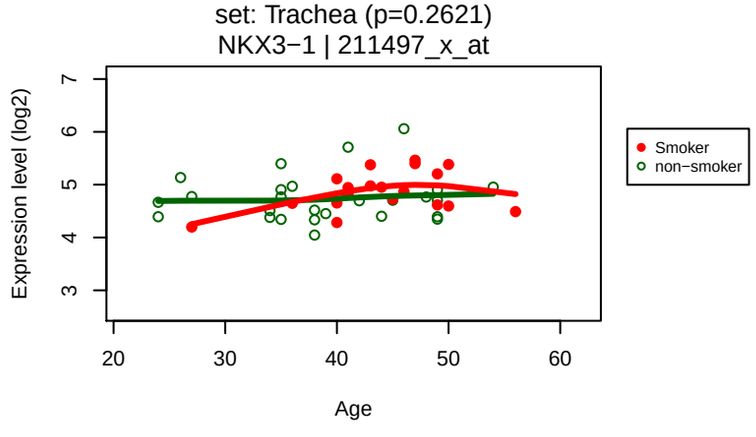

Gene expression ratio in Males
gene: NKX3−1 | 211497_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0931
p=0.592
p=0.238
p=0.159
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NKX3−1 | 211497_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4113
p=0.2949
p=0.7659
p=0.7319
SmallAirway2
LargeAirway2
Trachea
Macrophage2

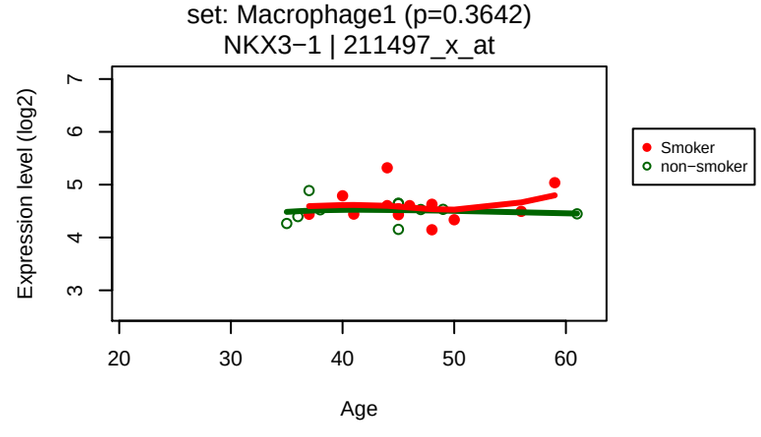

set: SmallAirway1 (p=0.0335)
NKX3−1 | 211497_x_at
Expression level (log2)
Age
Smoker
non−smoker

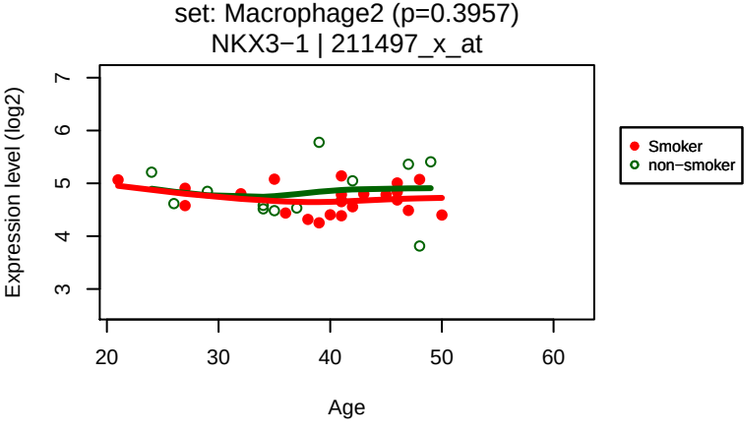

set: SmallAirway2 (p=0.1021)
NKX3−1 | 211497_x_at
Expression level (log2)
Age
Smoker
non−smoker

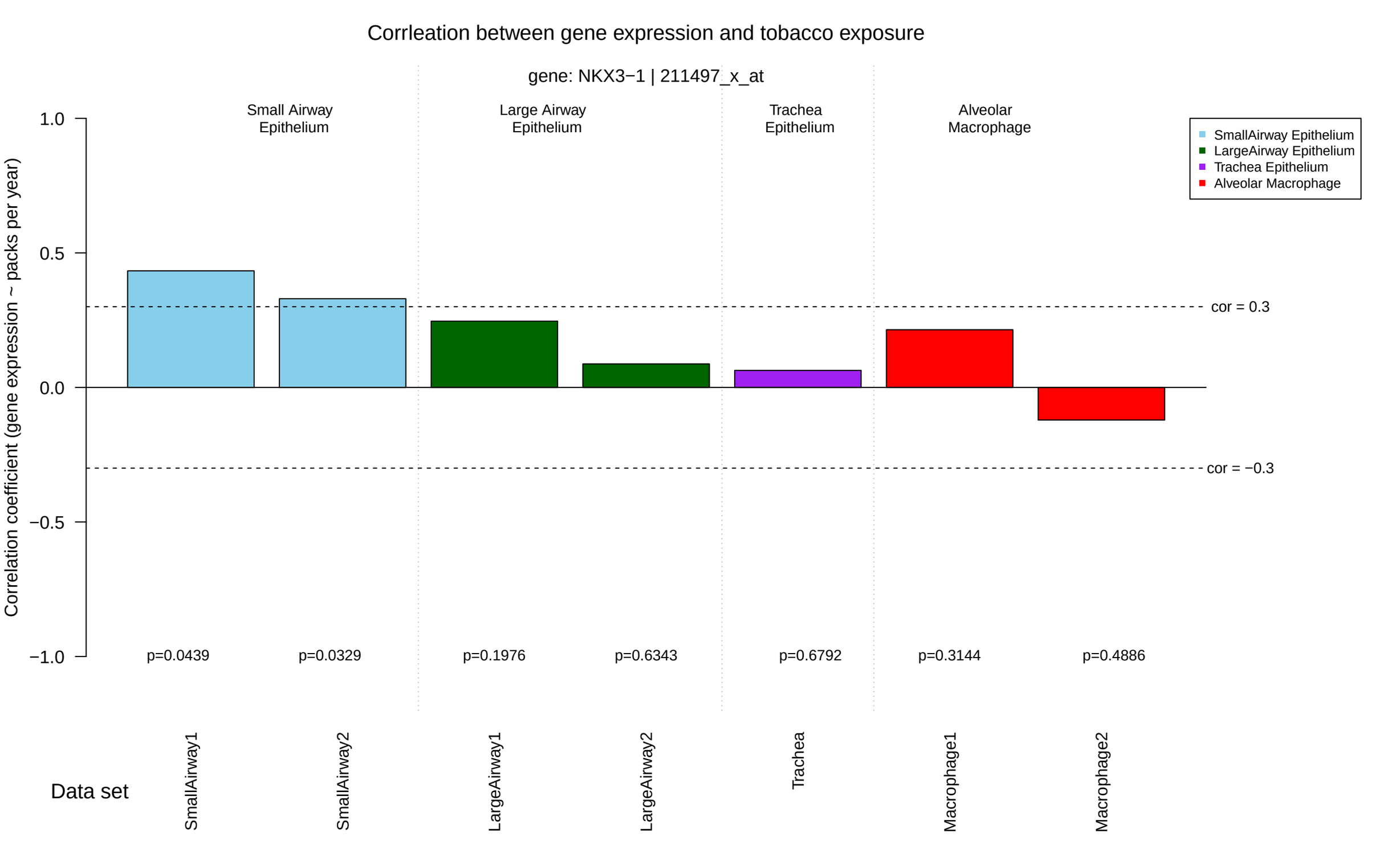

set: LargeAirway1 (p=0.0039)
NKX3−1 | 211497_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2219)
NKX3−1 | 211497_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2621)
NKX3−1 | 211497_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3642)
NKX3−1 | 211497_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3957)
NKX3−1 | 211497_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NKX3−1 | 211497_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0439
p=0.0329
p=0.1976
p=0.6343
p=0.6792
p=0.3144
p=0.4886
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2